다양한 센서를 이용하여 무선으로 생체 신호 측정이 가능한 초소형,저전력 생체 이식형 기기 설계

2021. 10. 06

박 홍 현

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지도교수 : 김상혁 교수님

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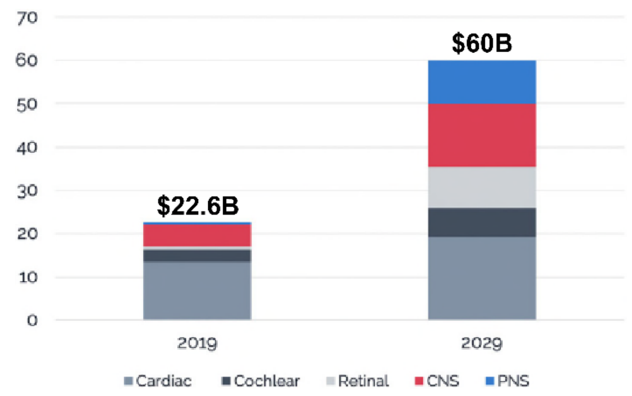
1. 시장 규모 파악 및 연구

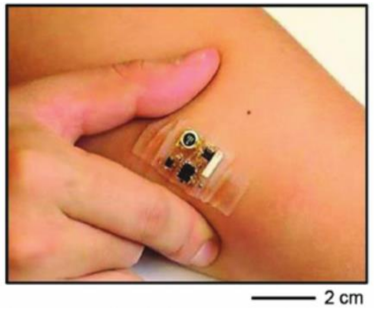
Bioelectronic Medicine Market Value, US$ Billions, 2019-2029



“Building a bioelectronic medicine movement 2019: insights from leaders in industry, academia, and research.” *Bioelectronic Medicine*, *6*(1).

웨어러블

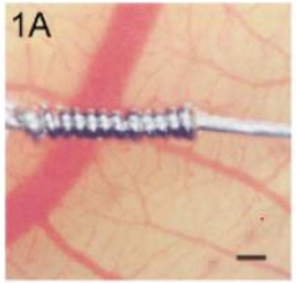
심전도 측정



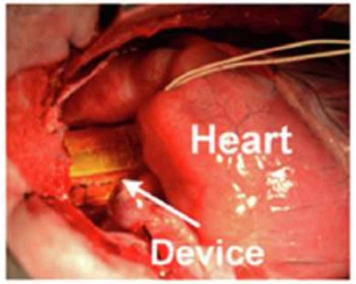
심전도 측정

Khan, Y., Ostfeld, A. E., Lochner, C. M., Pierre, A., & Arias, A. C. (2016). Monitoring of Vital Signs with Flexible and Wearable Medical Devices. Advanced Materials, 28(22), 4373–4395.

삽입형



혈당 측정

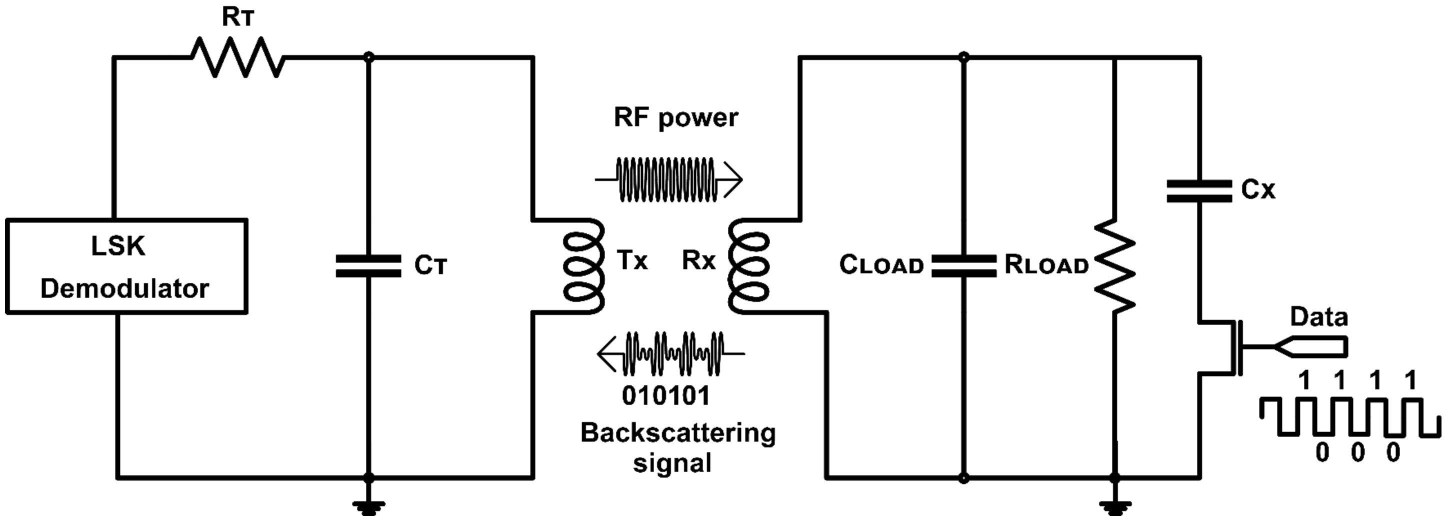


혈압 측정

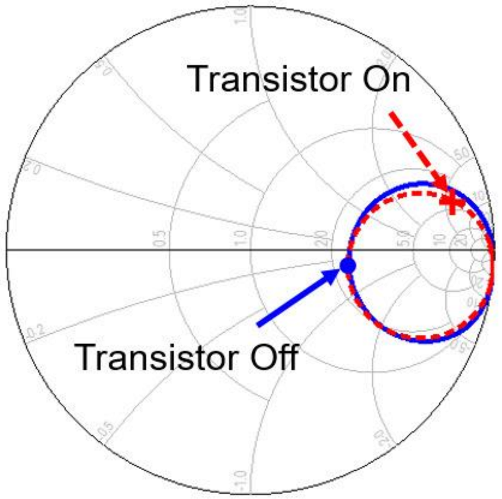
Koschwanez, H. E., & Reichert, W. M. (2007). In vitro, in vivo and post explantation testing of glucose-detecting biosensors: Current methods and recommendations. Biomaterials, 28(25), 3687–3703.

Zhang, H., Zhang, X.-S., Cheng, X., Liu, Y., Han, M., Xue, X., … Xu, Z. (2015). A flexible and implantable piezoelectric generator harvesting energy from the pulsation of ascending aorta: in vitro and in vivo studies. Nano Energy, 12, 296–304.

2. Load-shift keying(LSK) modulation



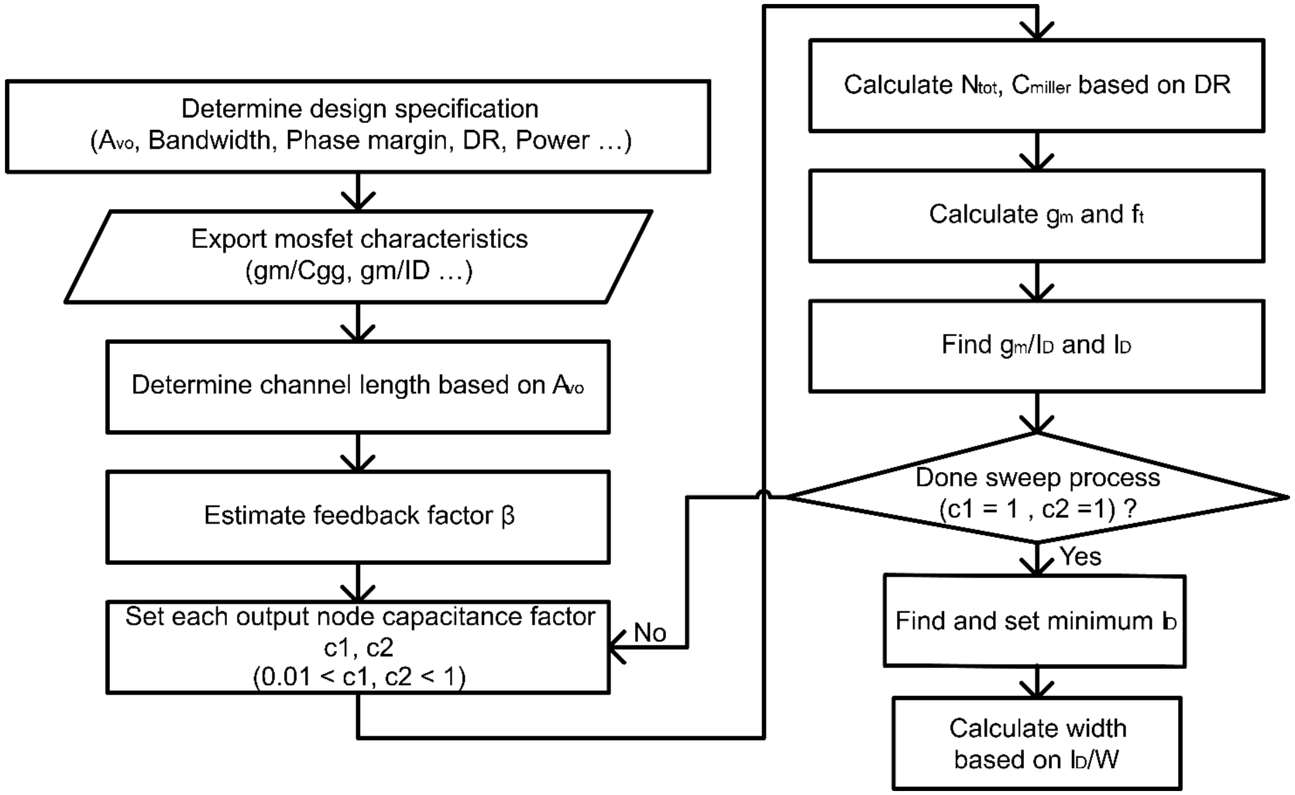
< Simple LSK modulation circuit >



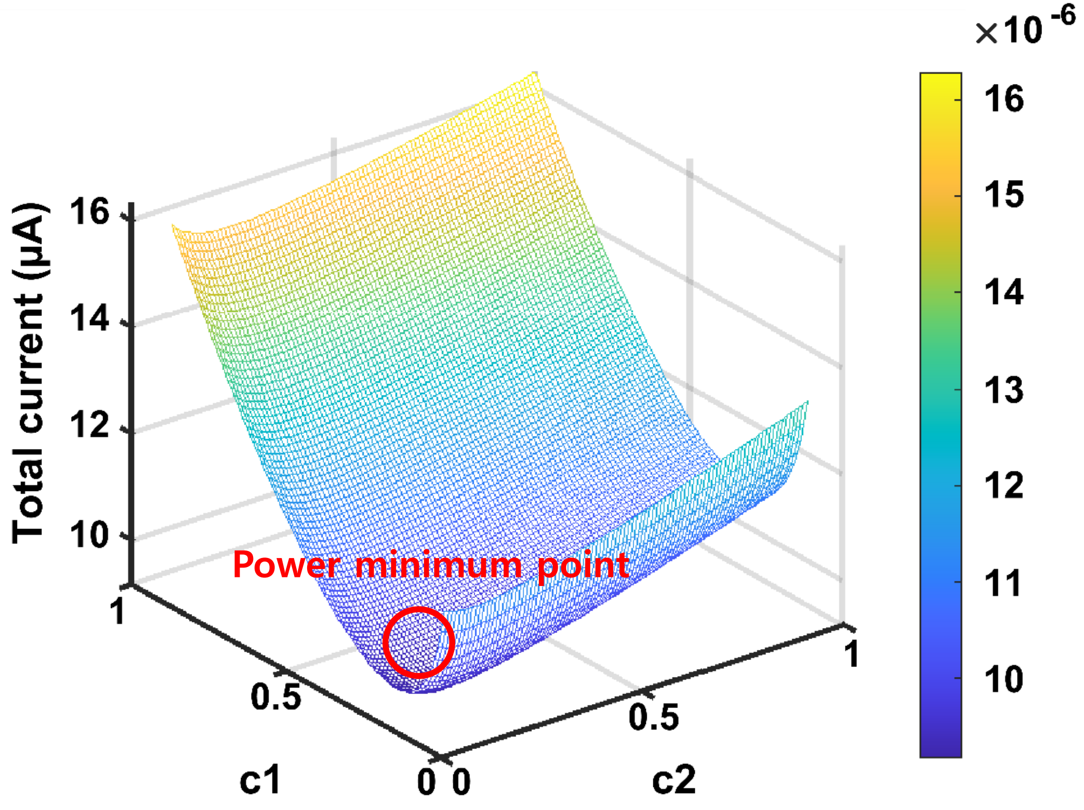
< Impedance switching result >

- 별도의 ADC를 필요로 하지 않음
- 전력 소모 감소
- Rx 코일과 Load 간의 conjugate matching 중요

2. gm/id based 증폭기 최적화



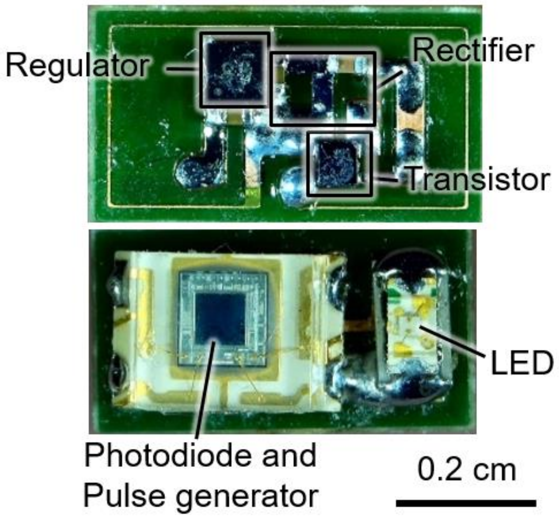
< 증폭기 최적화 플로우 차트 >



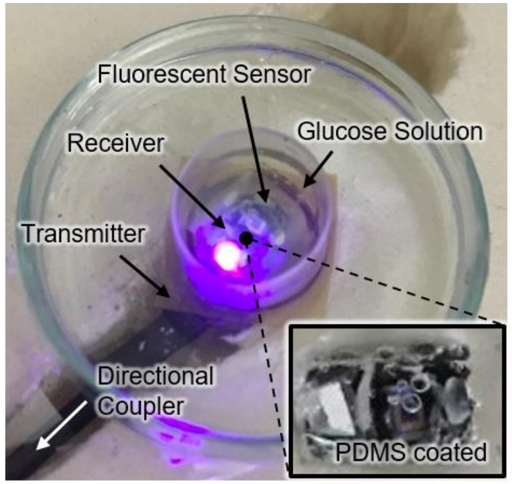
< 증폭기 최적화 결과>

Boris Murmann, “Stanford University EE214 Lecture Notes”, https://picture.iczhiku.com/resource/eetop/shIfgiaFuPGIOmXm.pdf

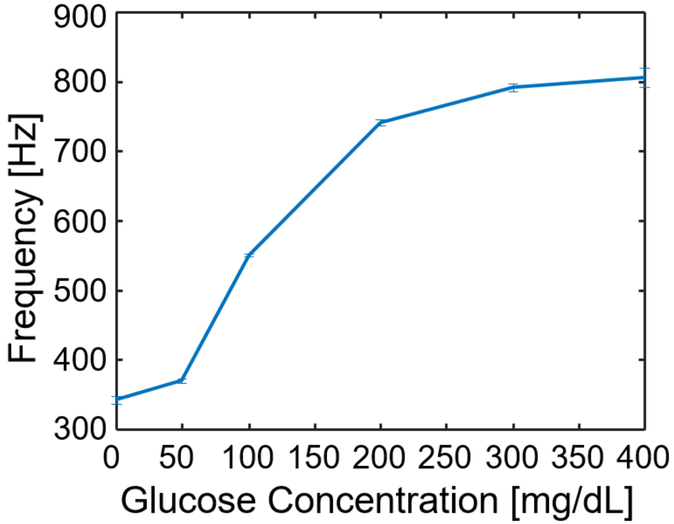
3. 형광 센서를 이용한 혈당 측정 기기 선행 연구



< 시제품 >



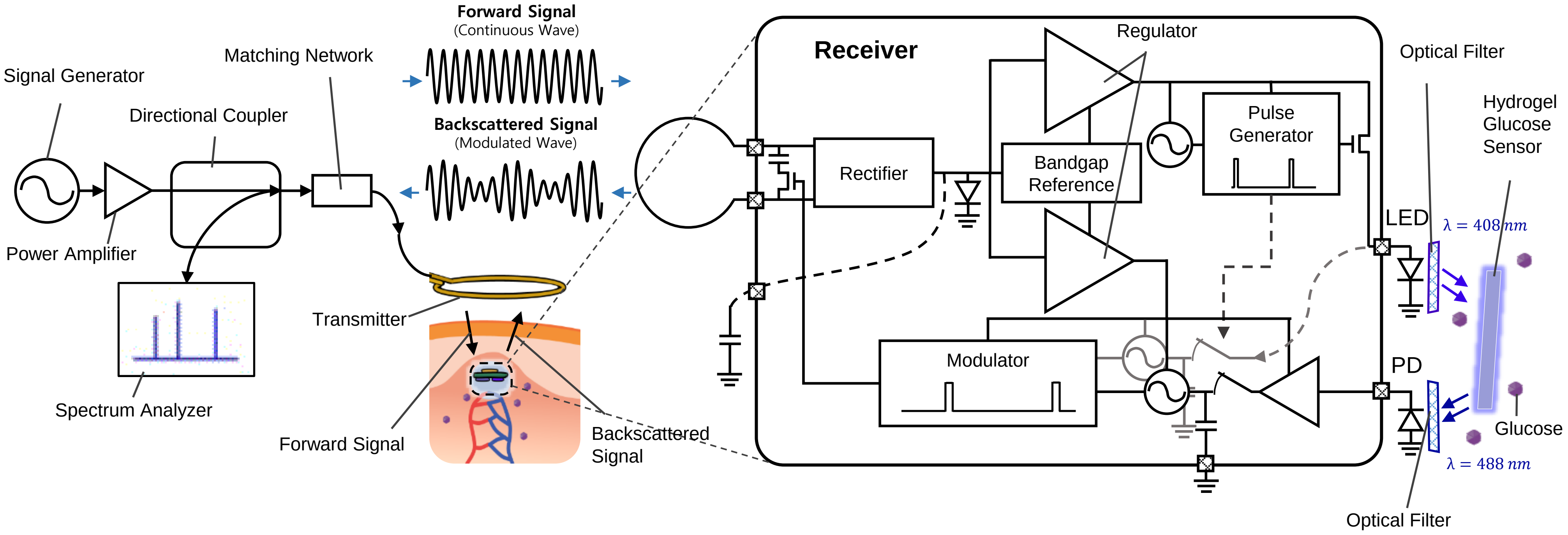
< 실험 환경 >



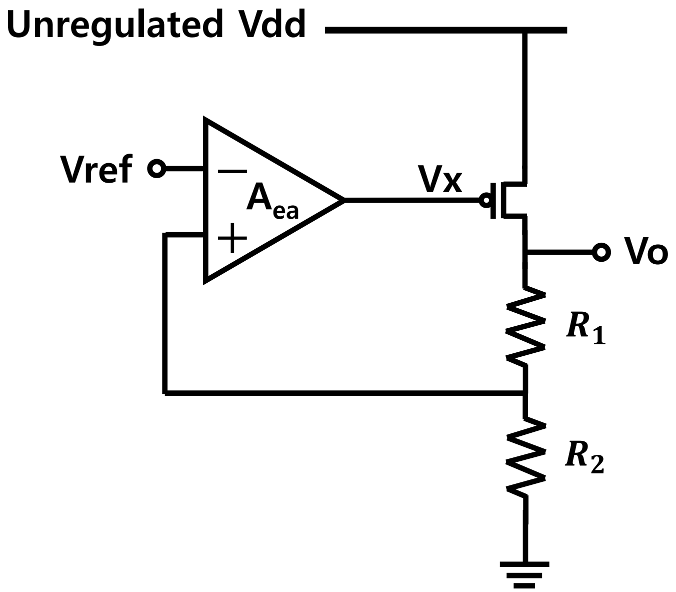
< 측정 결과 >

- **낮은 무선전력전송 효율 (η = 0.75 %)**
- **LED에서 지속적인 전력 소모 (15 mW)**
- **주사 바늘을 통해 이식하기에 부담되는 크기 (3 mm x 6 mm)**

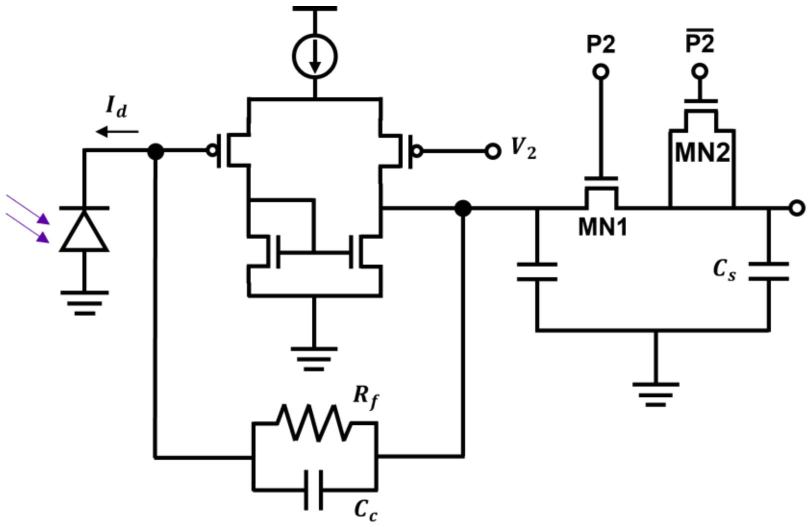
3. 시스템 블록 다이어그램



3. 핵심 회로도 (레귤레이터, TIA)



< LDO 레귤레이터>

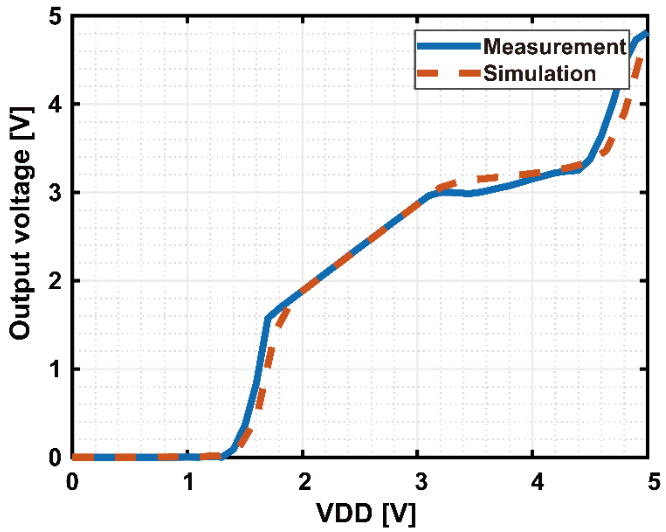


< Transimpedance amplifier (TIA) >

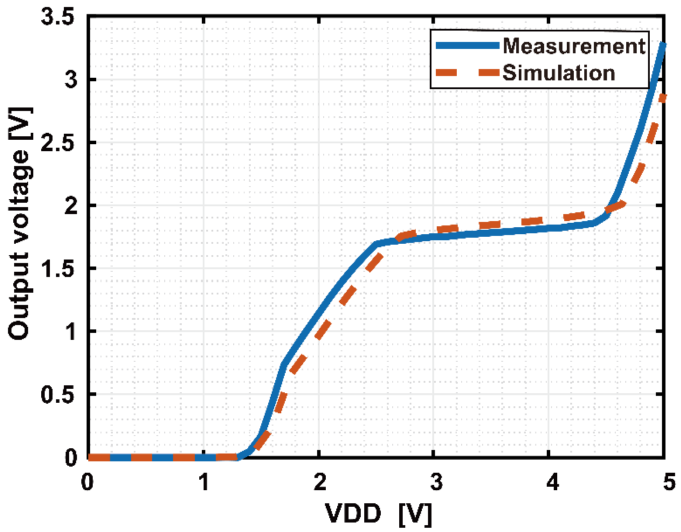
$$Vo = \frac{V_{ref}}{A_{ea}} + \frac{V_{ref}}{\beta}, \quad \beta = \frac{R_2}{R_1+R_2}$$

$$Vo = R_f * I_d$$

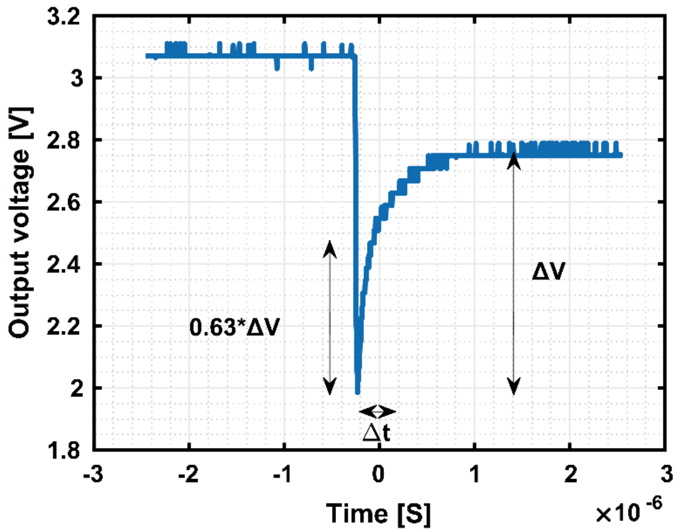
3. 측정 결과 (LDO 레귤레이터)



< LDO for LED >

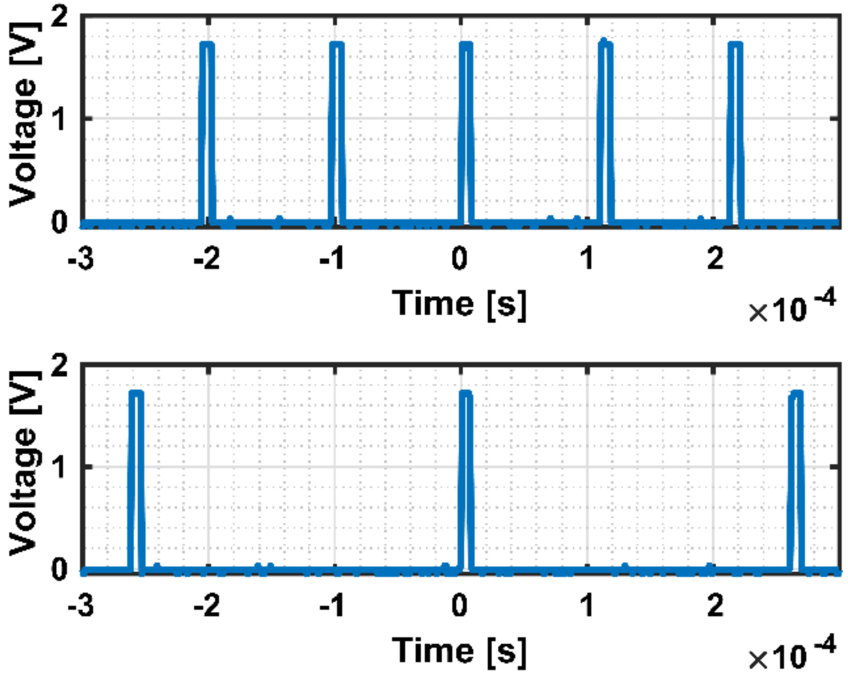


< LDO for sensor module >

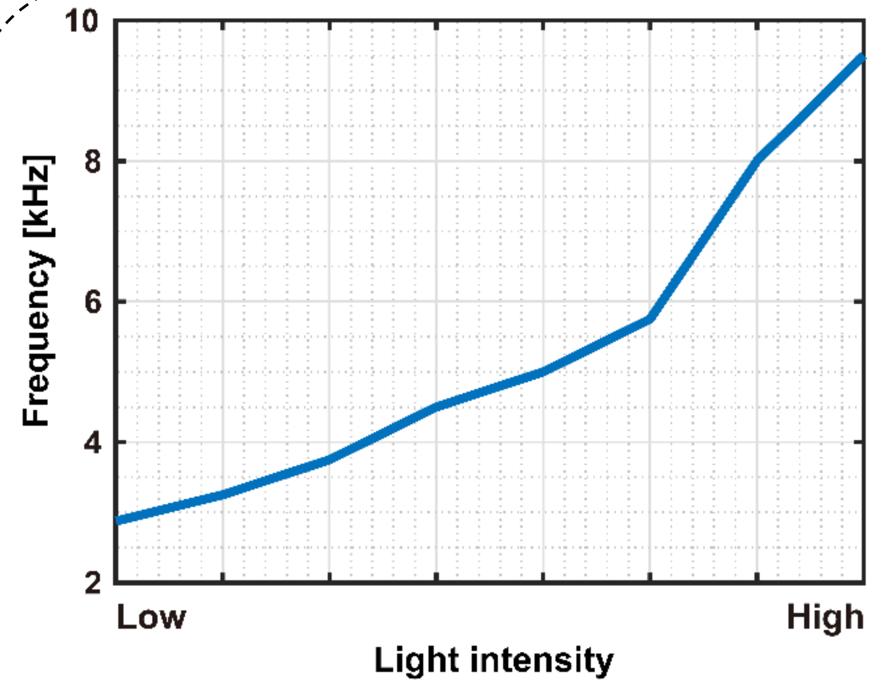


< Undershoot 전압 회복 결과 >

3. 측정 결과 (센서 모듈)

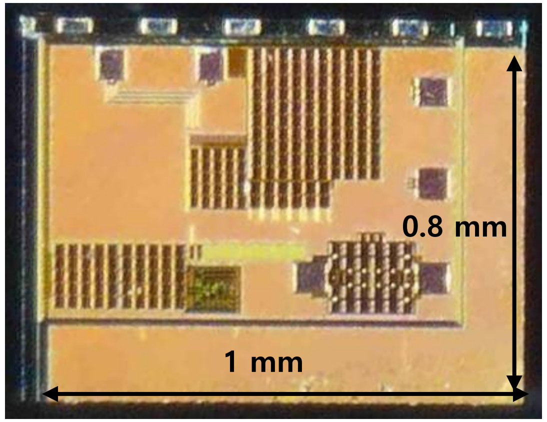


< 빛의 세기에 따른 Time-domain 결과 >

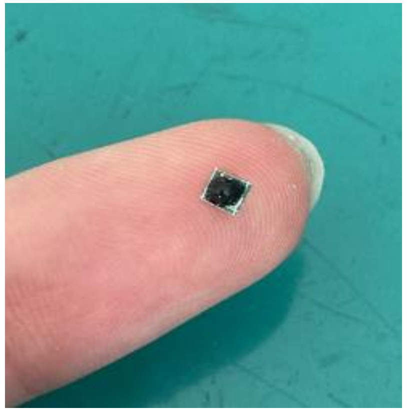


< 빛 세기에 따른 주파수 변화 범위>

3. 연구 요약



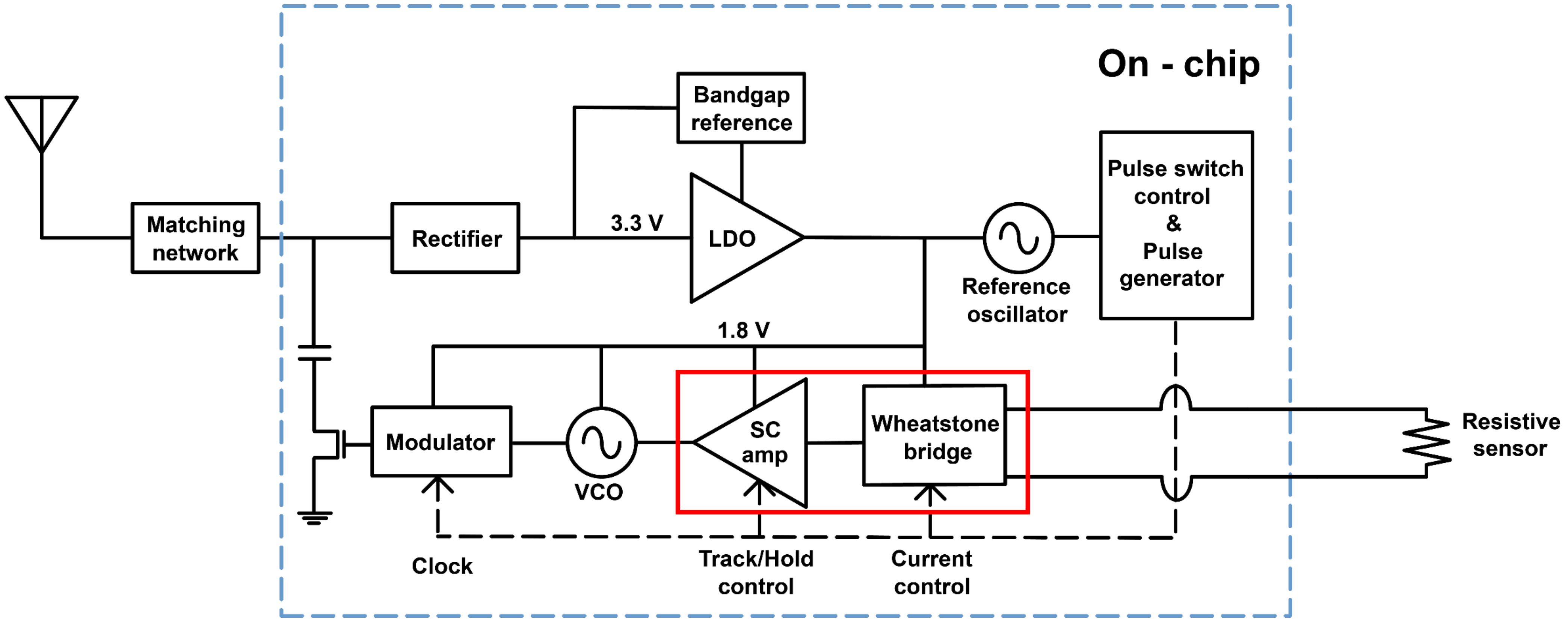
< Chip photomicrograph >



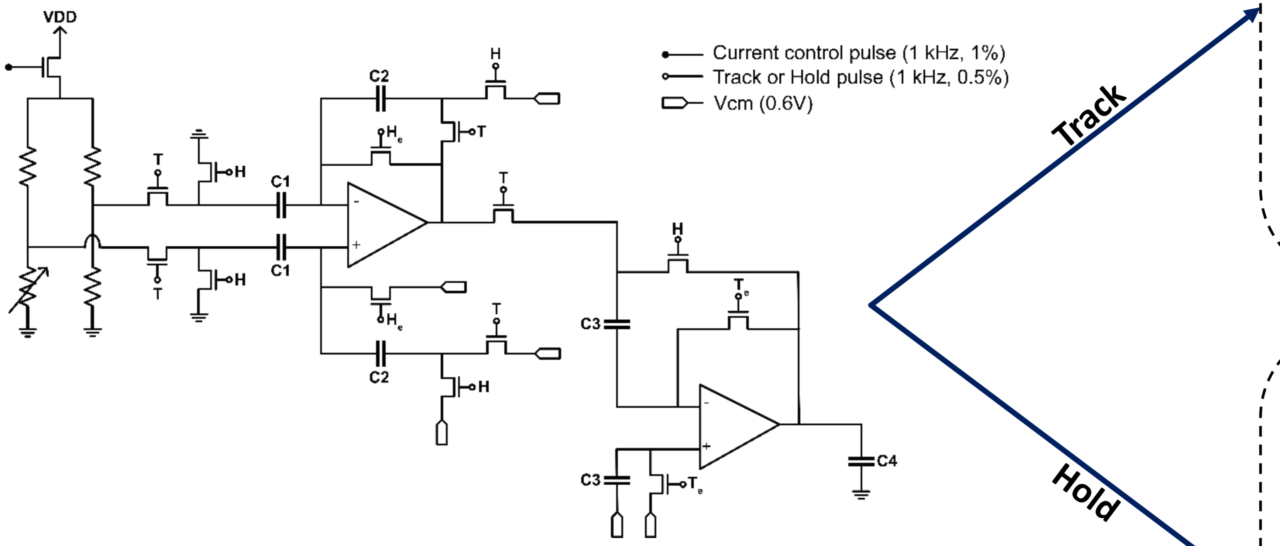
< Glucose readout sensor image >

	선행 연구	현행 연구
변조 방식	LSK(Load-shift keying)	LSK(Load-shift keying)
공정	Discrete component	TSMC 180nm MS
Bare chip 크기	.	1 mm x 0.8 mm
이식 기기 크기	6 mm x 3 mm	2.5 mm x 2.6 mm
소모 전력	15 mW	150 uW
Carrier 주파수	1.5 GHz	1.2 GHz
Signal 주파수	350 kHz ~ 850 kHz	2.6 kHz ~ 9.5 kHz

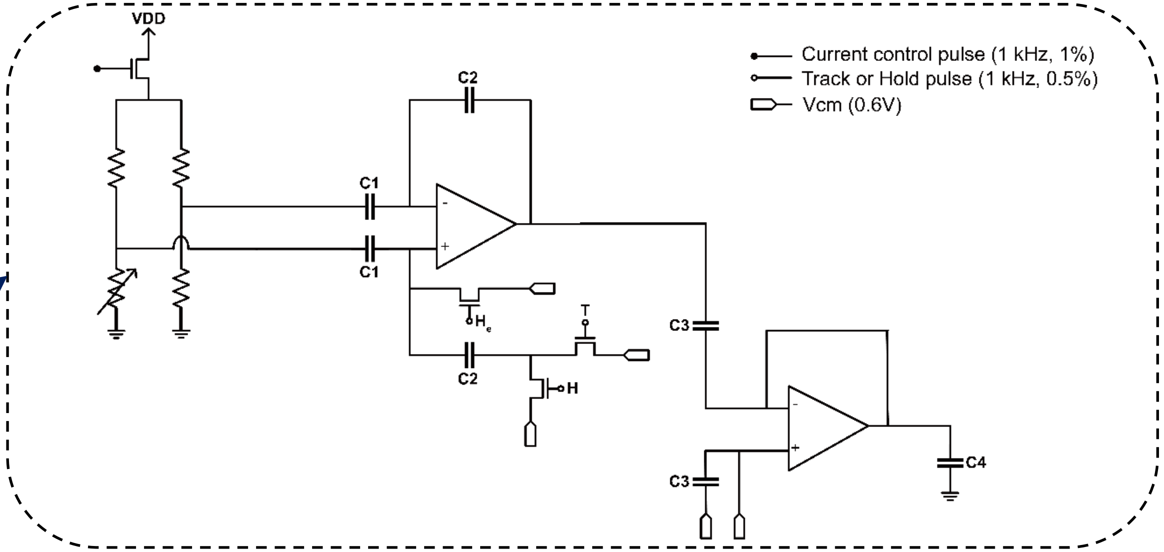
4. 시스템 블록 다이어그램

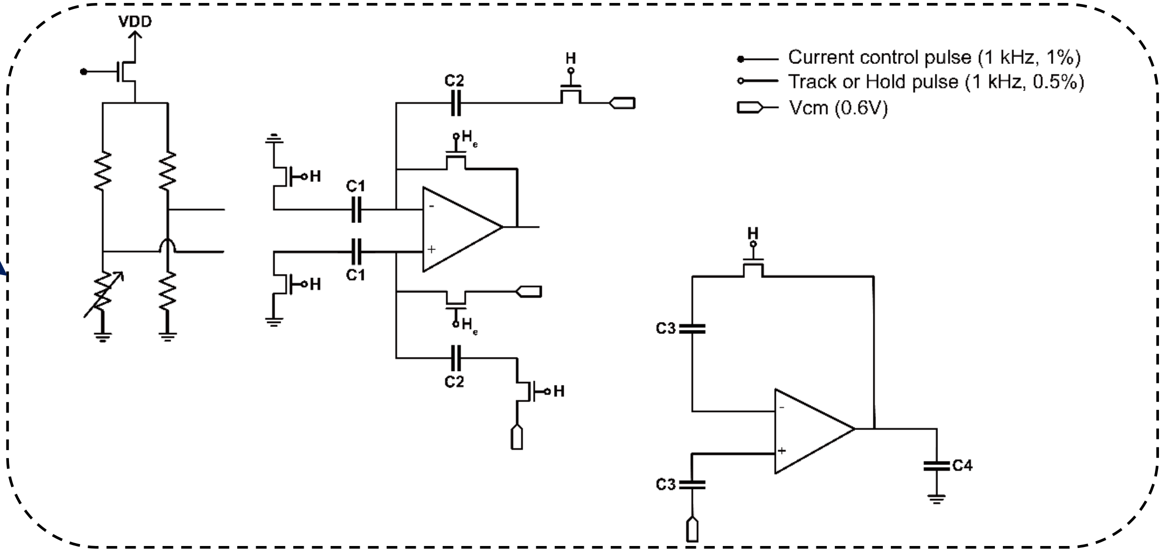


4. 생체 신호 증폭기 회로도

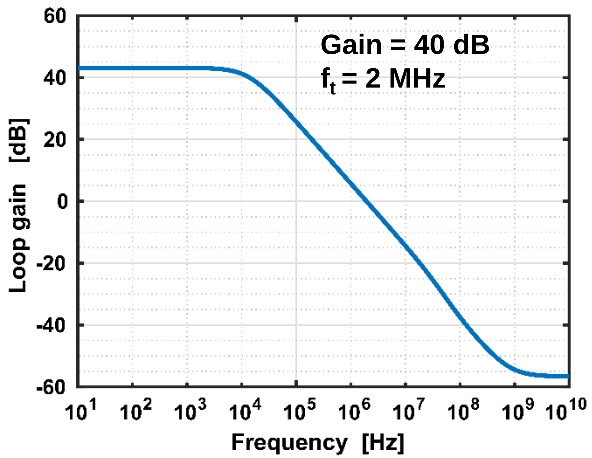


< SC resettable gain amplifier >

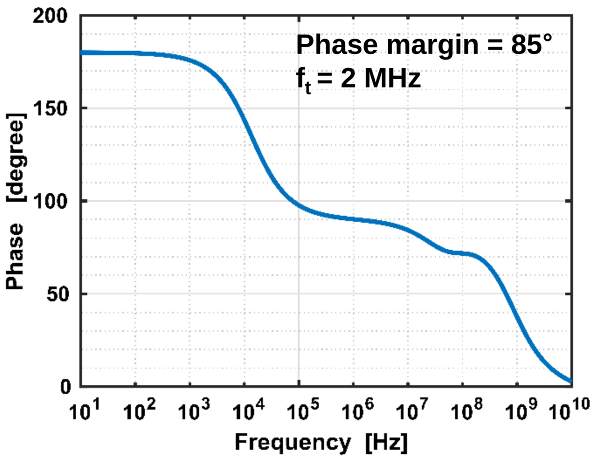




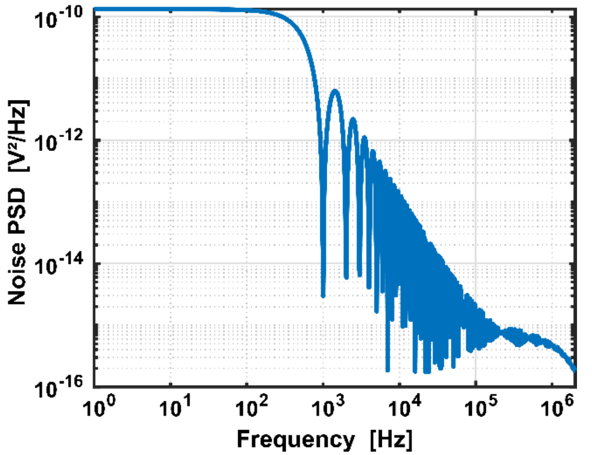
4. 증폭기 시뮬레이션 결과



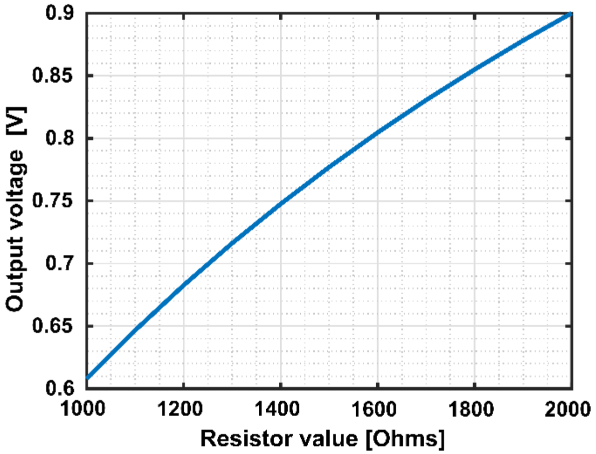
< Loop gain bode plot >



< 위상 bode plot >



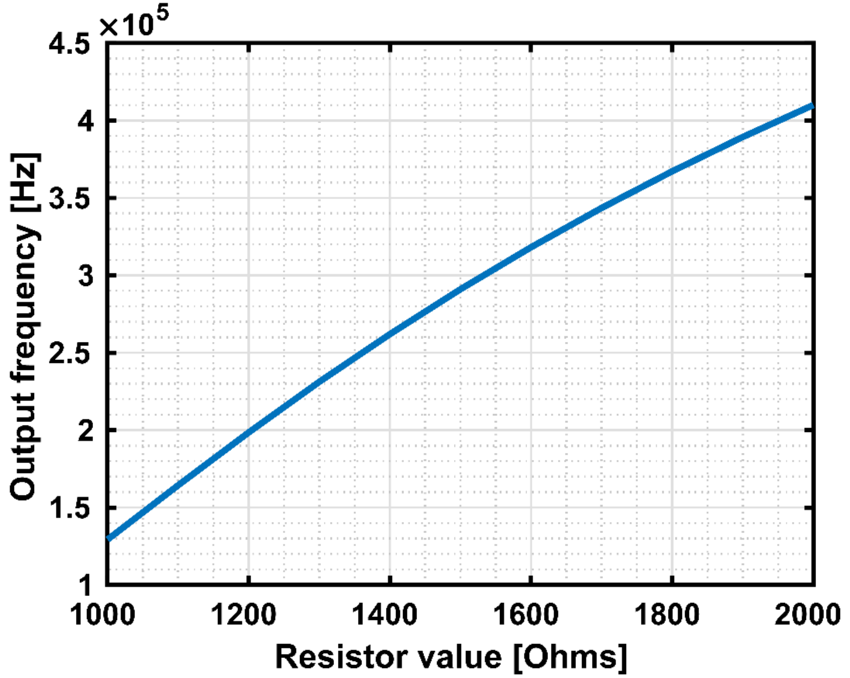
< T/H 잡음 결과 >



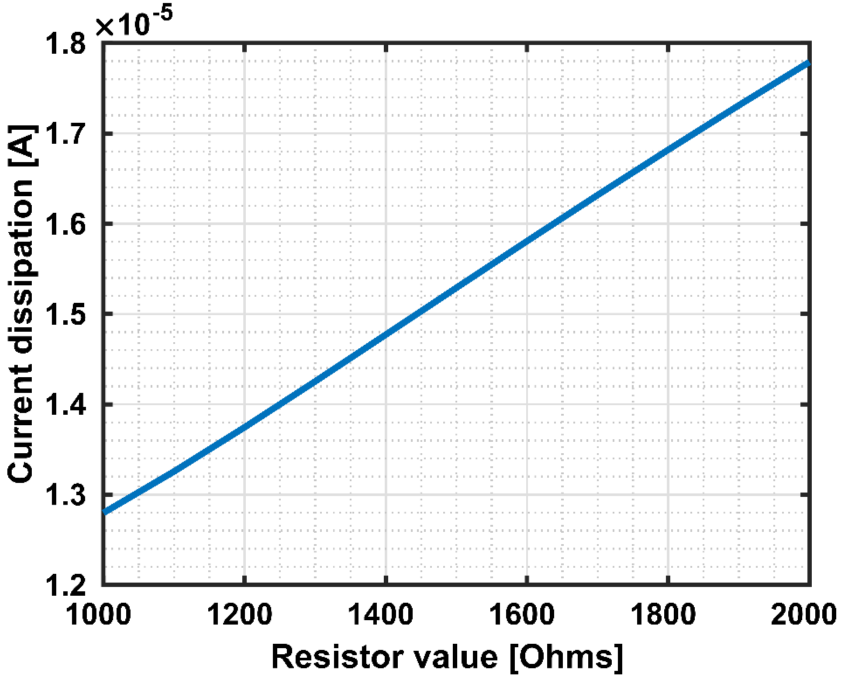
< Resistor vs Output voltage >

Loop gain	40 dB
Phase margin	89°
Bandwidth	1.98 MHz
Noise (gain stage)	$1.37 * 10^{-5}$
Noise (Buffer)	$2.32 * 10^{-9}$
Dynamic range (DR)	36 dB
소모 전력	18.5 uW

4. 생체 신호 판독 모듈 결과

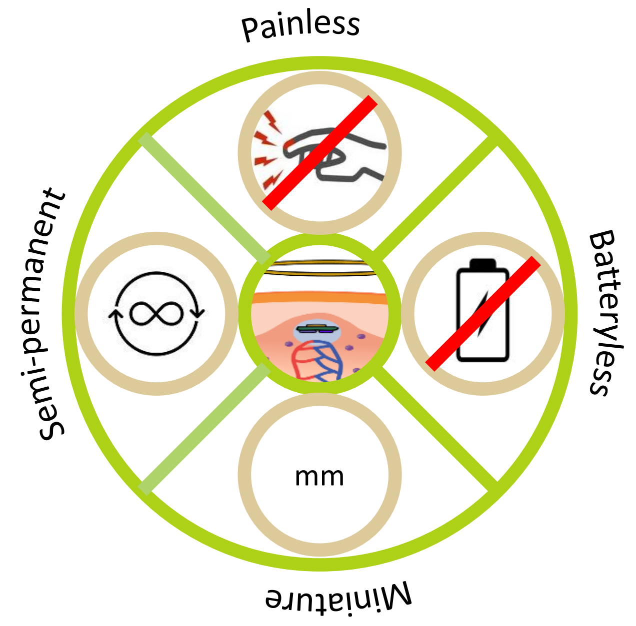


< frequency range by resistance>



< Current consumption by resistance>

5. 결론



- Fluorescent/Strain sensor와 결합하여 생체 신호(혈관 부피, 혈당)측정이 가능한 이식기기 설계
- 무선 전력 전송을 이용하여 기기가 소모하는 전력을 공급하고 부피가 큰 배터리를 제거
- Load-shift keying(LSK) 변조를 통해 생체 신호를 체외로 전송
- 생체에 이식이 가능하도록 저전력 및 초소형 크기 설계
- 반영구적으로 사용 가능하며 연속적인 생체 신호 측정